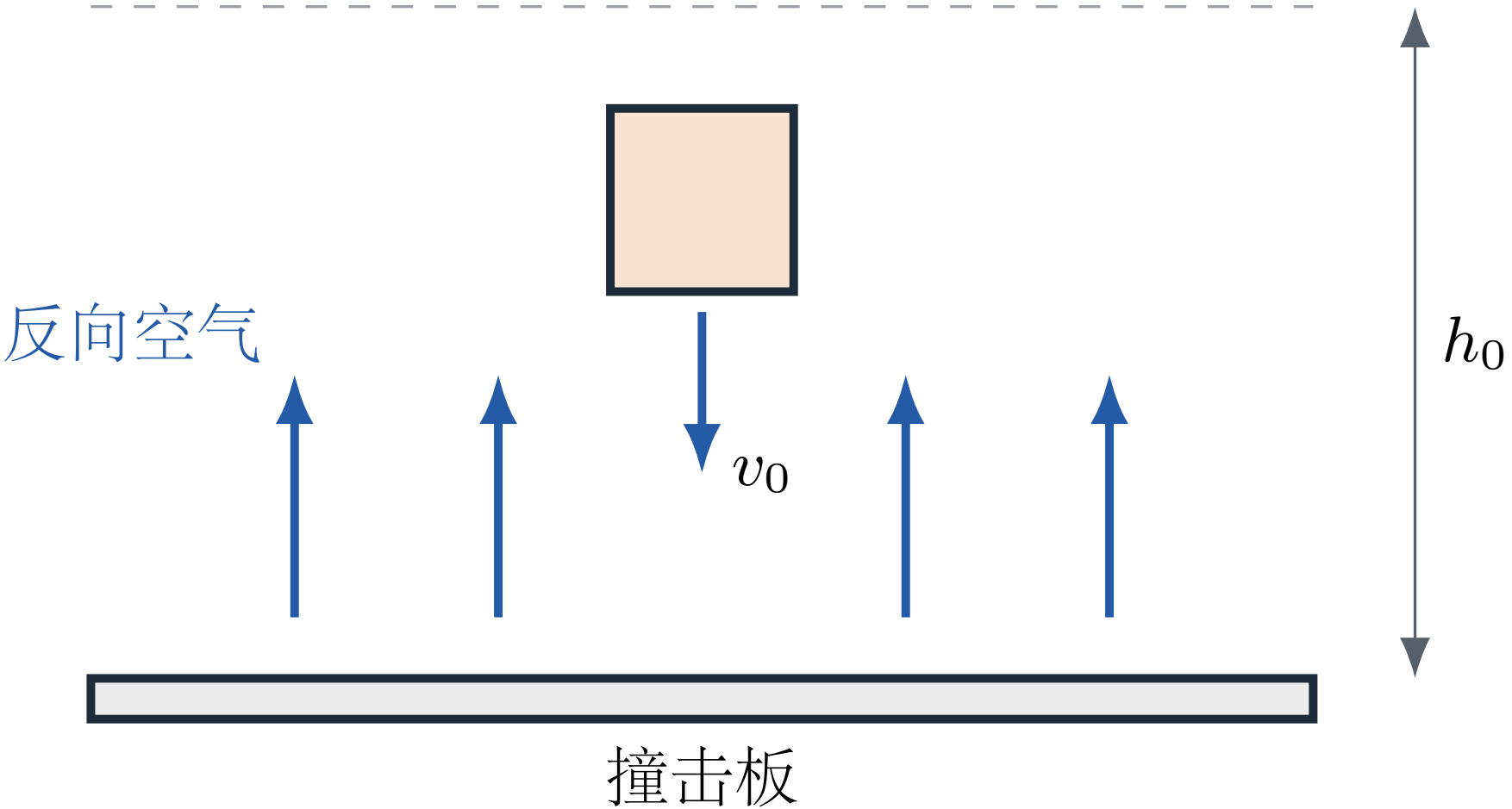
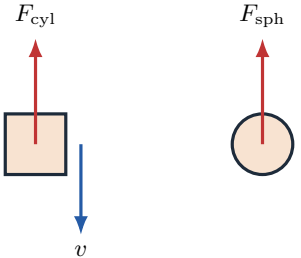
颗粒仍在空气层内时受到阻力；图为竖直截面



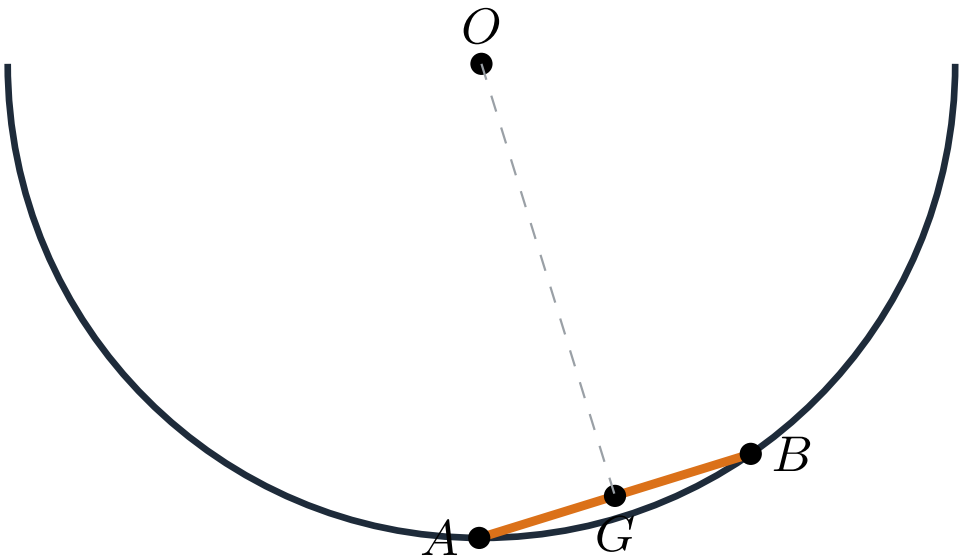


相对速度 $U = v_0 + v$ 随颗粒减速而改变

$$v\,dv/dz = -\eta\rho_0(v_0+v)^2/(\rho r)$$

$$s = \rho r(\ln 2 - 1/2)/(\eta\rho_0)$$

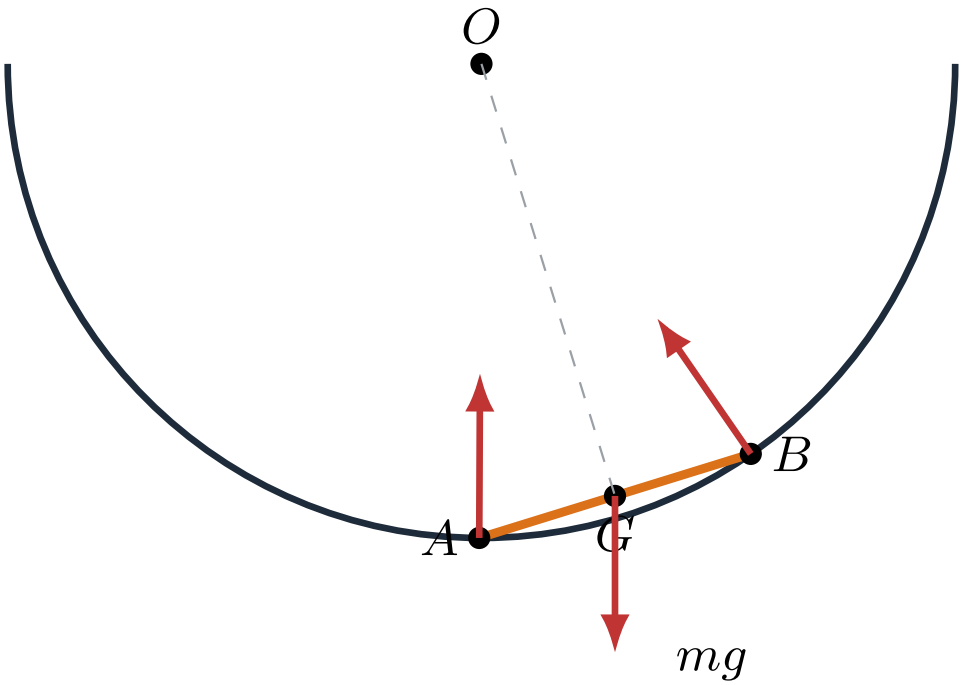
恒定 $U = 2v_0$ 另列为初值估算



$R > 1.5L$

均匀杆长为 L

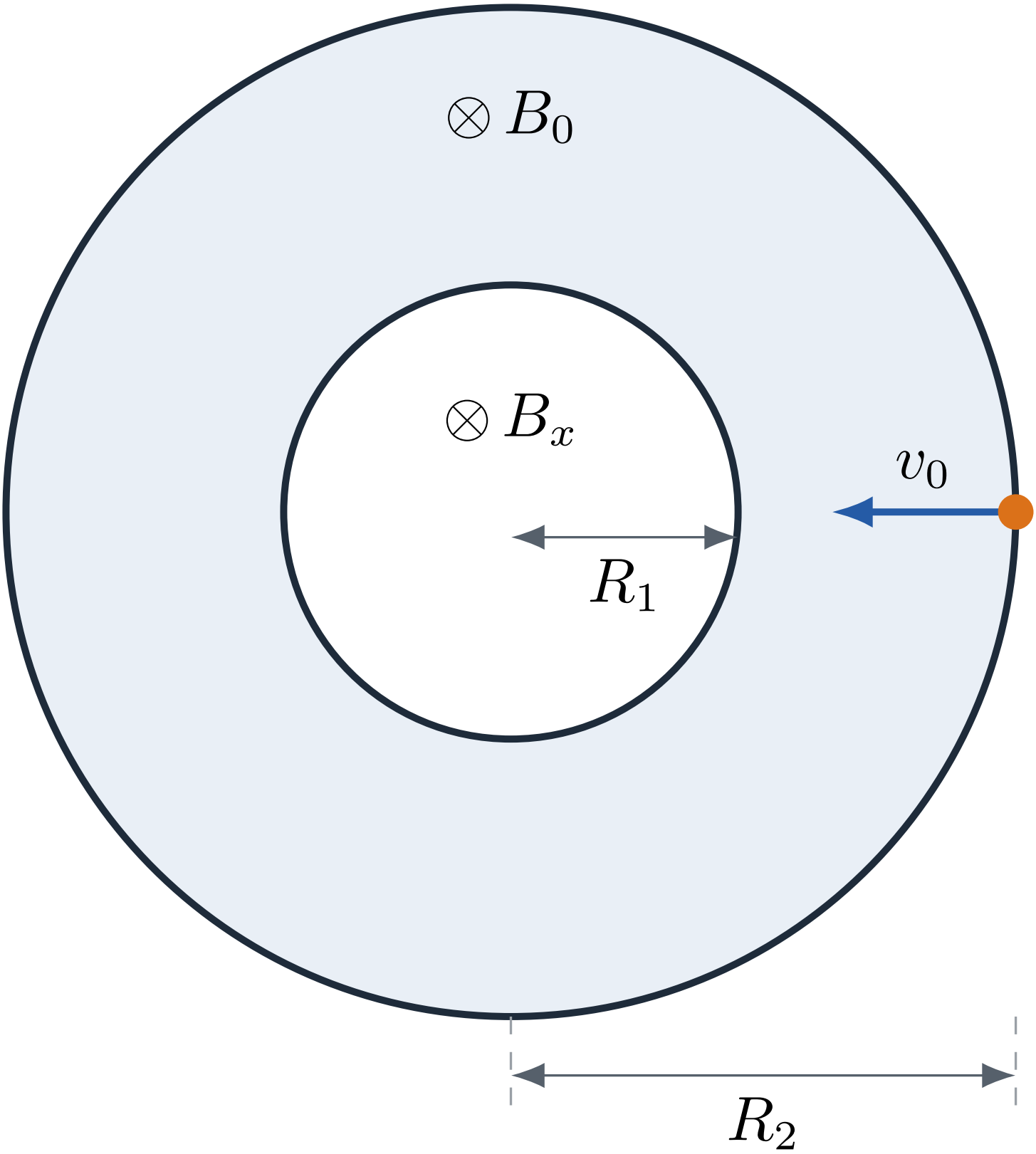
两端均接触圆周，杆与 OG 垂直

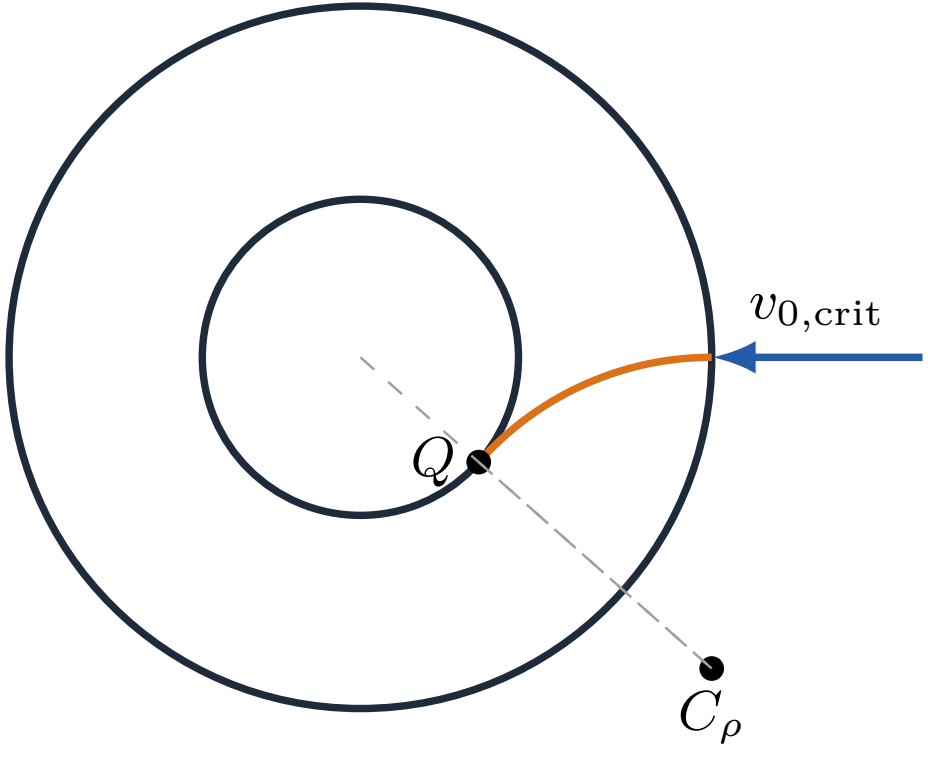


$$d = \sqrt{R^2 - L^2/4}$$

$$K = d^2 + L^2/12$$

两端均接触圆周，杆与 OG 垂直


$\otimes B_0$
$\otimes B_x$
v_0
R_1
R_2

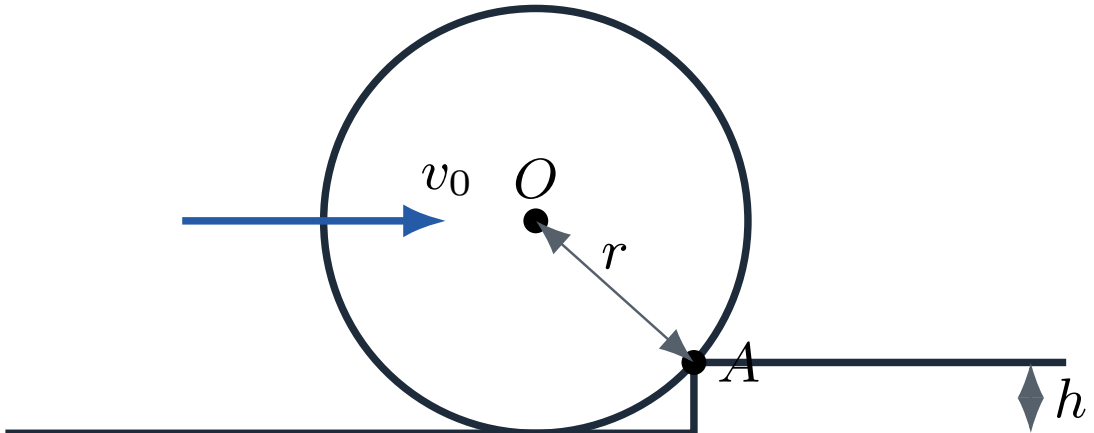


$$\rho = (R_2^2 - R_1^2)/(2R_1)$$

$$|OC_\rho| = R_1 + \rho$$

临界轨道在 Q 与内圆相切

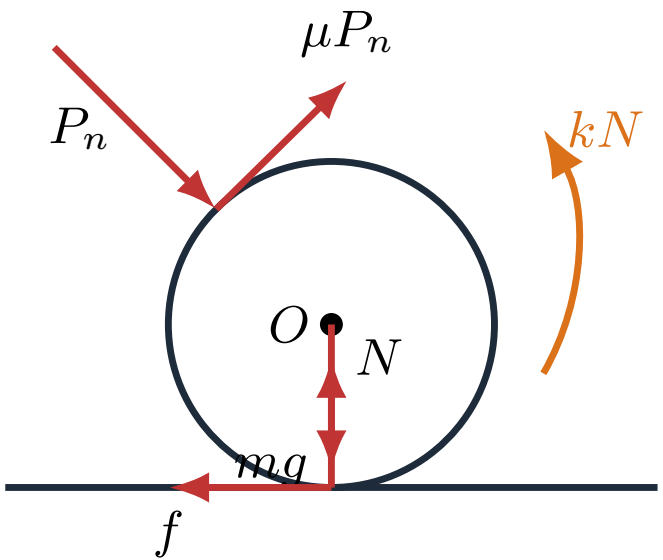
真正进入内圆须 $v_0 > v_{0,\mathrm{crit}}$



碰撞后绕台阶角 A 转动

碰前圆环向右纯滚动

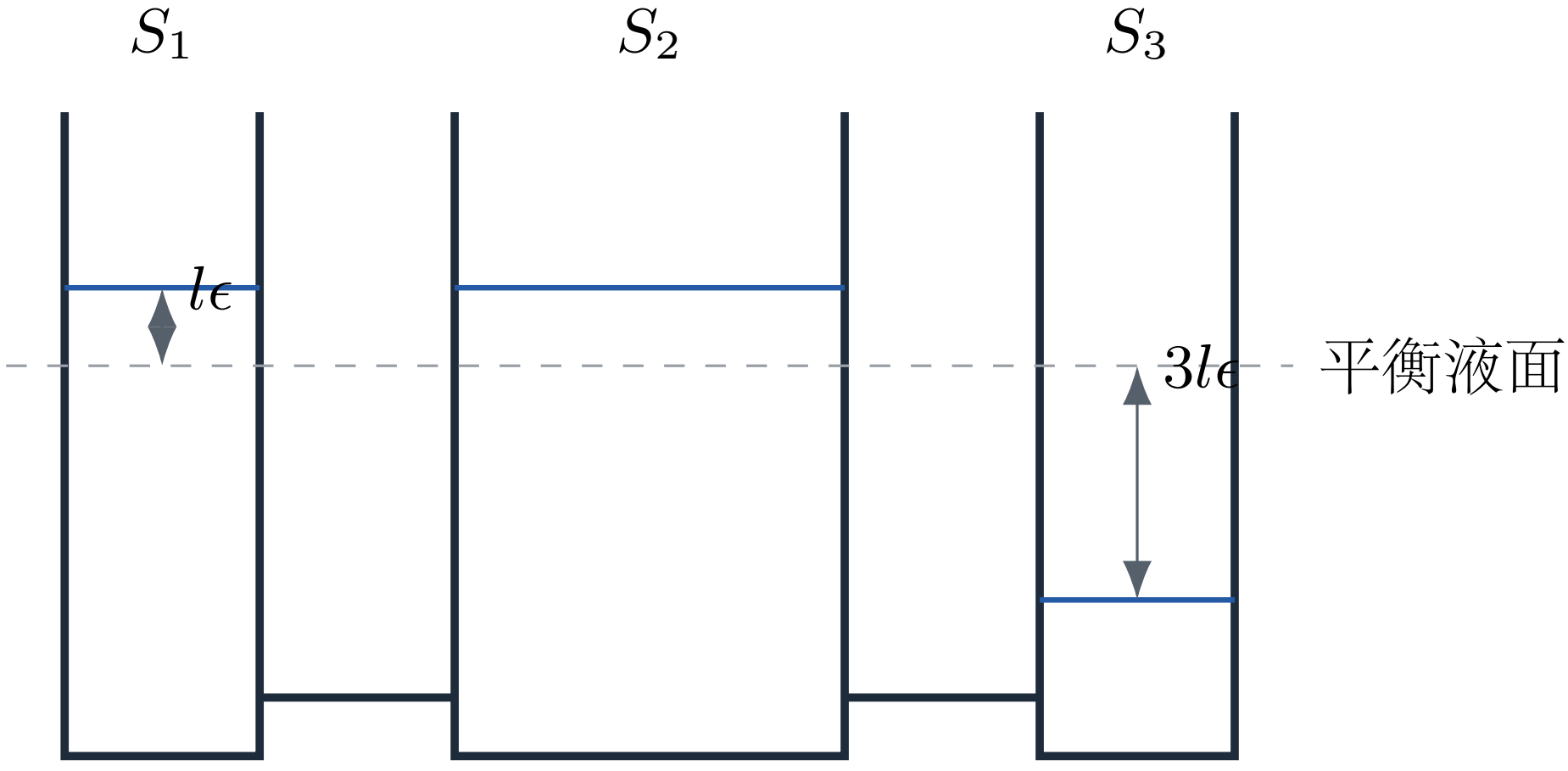
越障条件与停止时间待求



受力点区分：重力在 O，摩擦在接触点

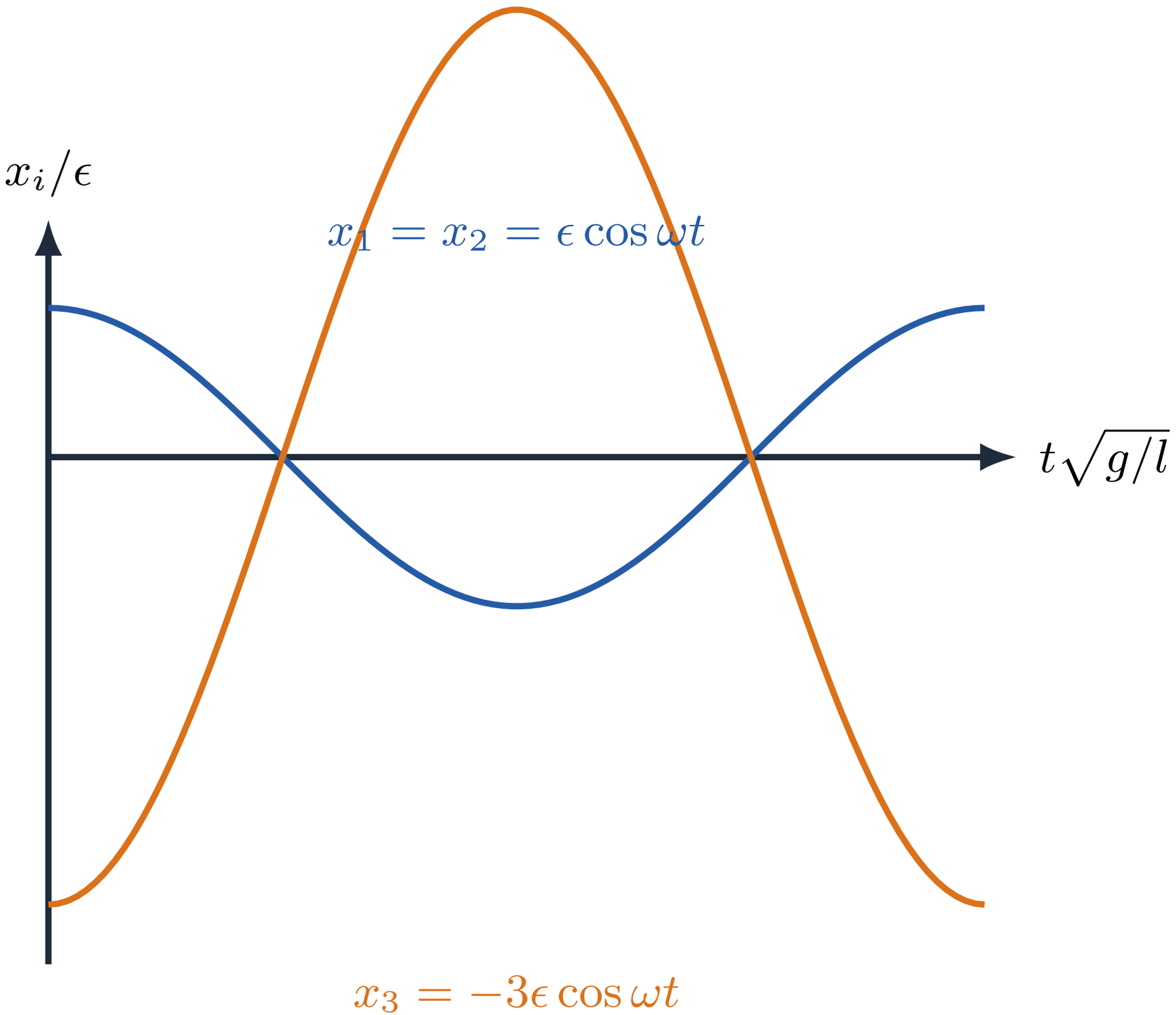
棍的法向与切向力互相垂直

滚动阻力偶矩与滚动角速度反向

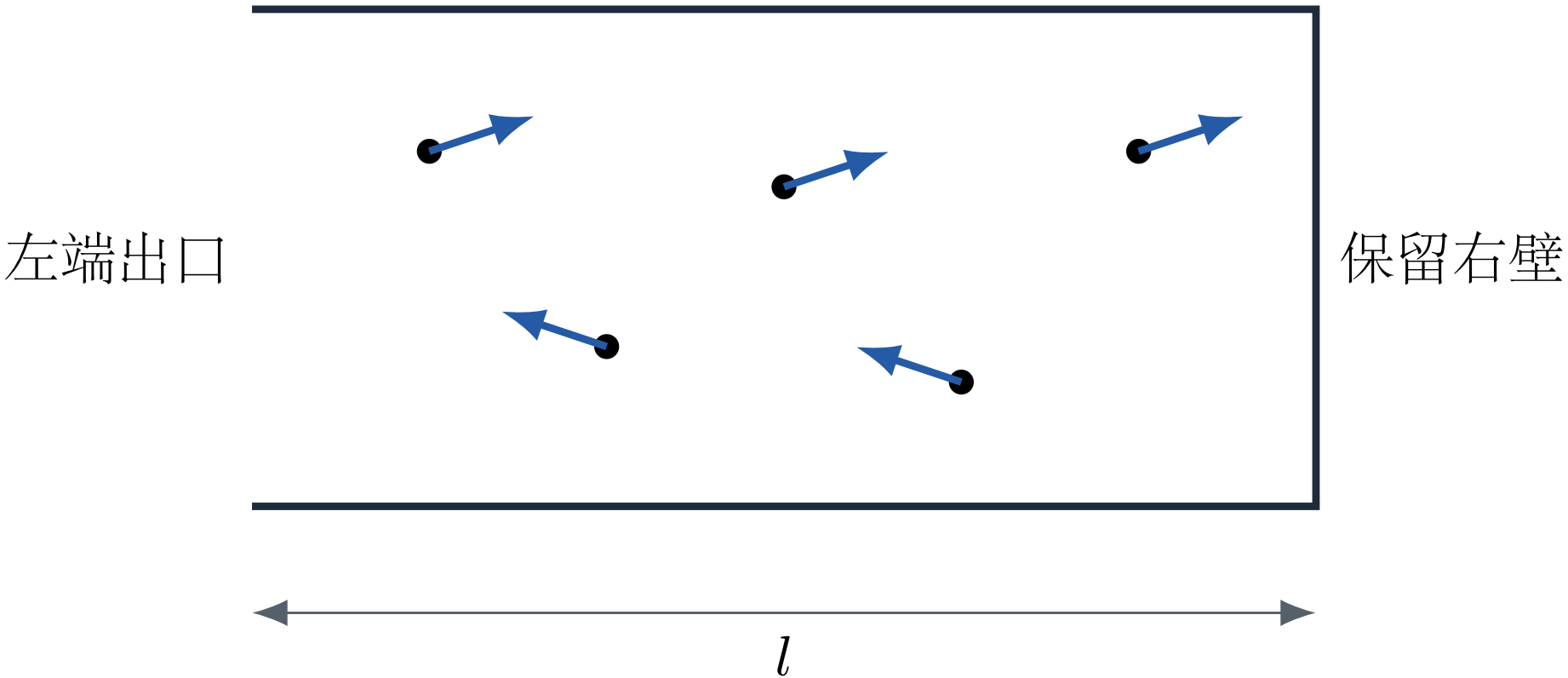


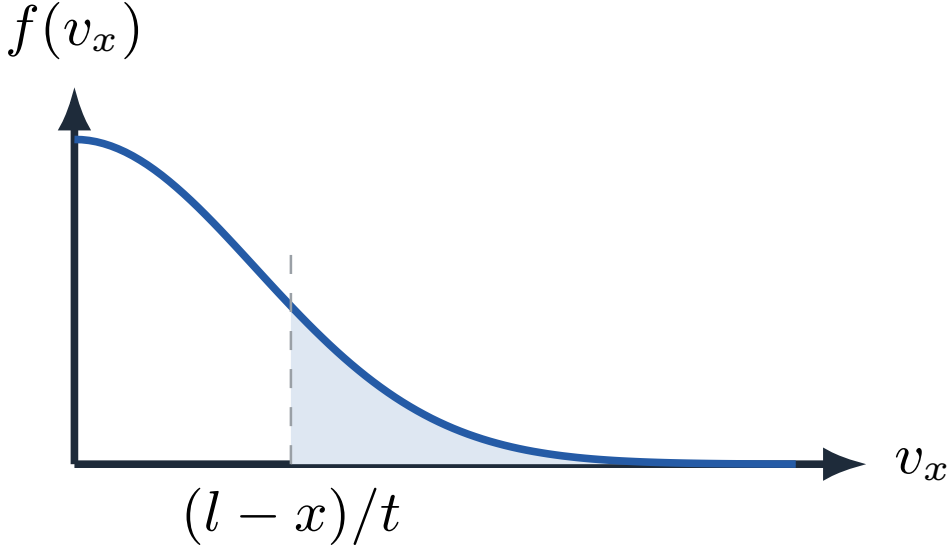
$S_1 : S_2 : S_3 = 1 : 2 : 1$; 位移定义为 lx_i

$\omega = \sqrt{g/l}$；两条曲线按振幅比例绘制



初始热平衡：分子速度既有向右也有向左



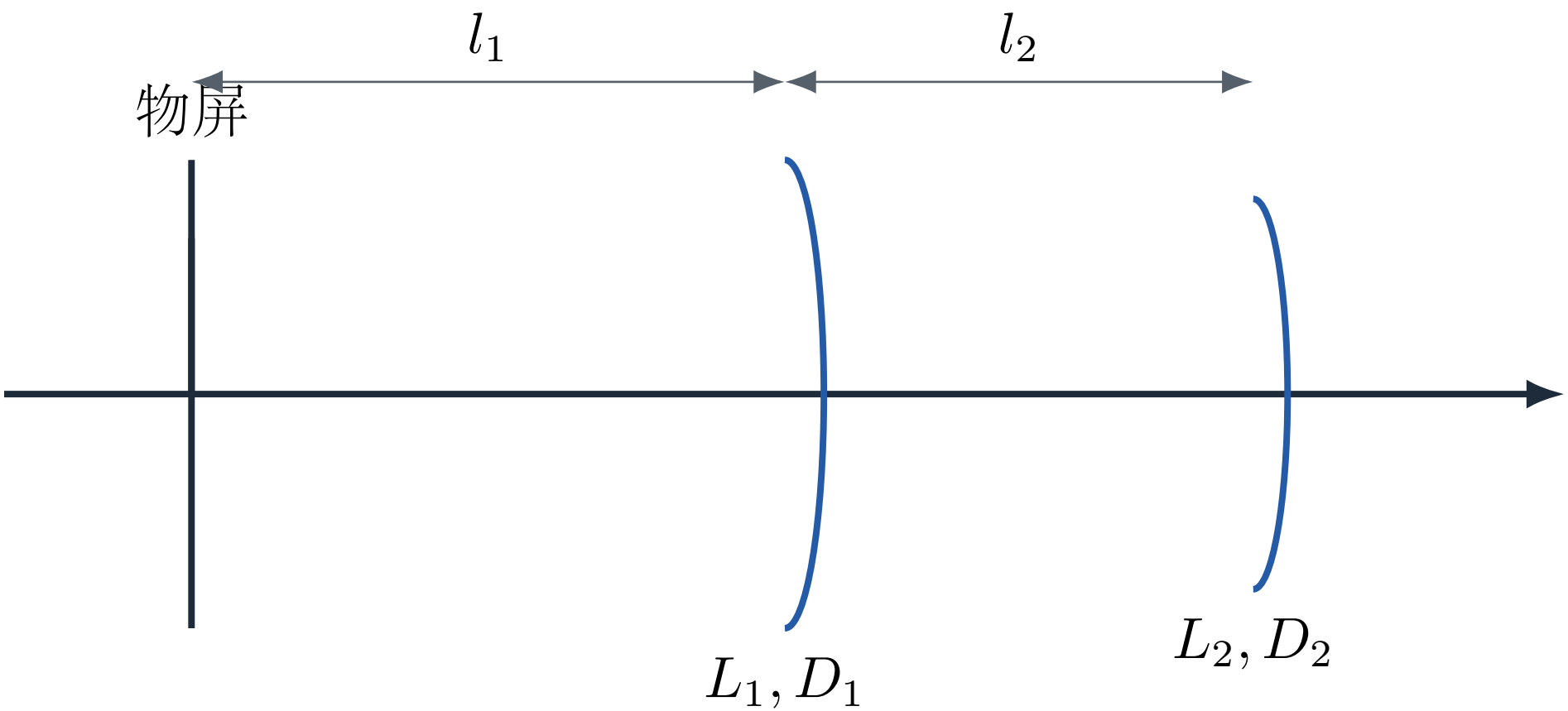


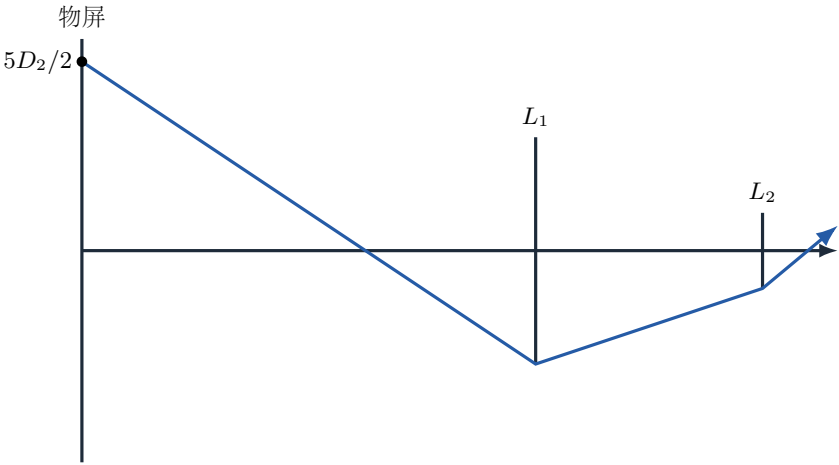
位置为 x、初速度向右的分子

撞右壁须 $v_x > (l-x)/t$

初速度向左者直接逸出，不撞右壁

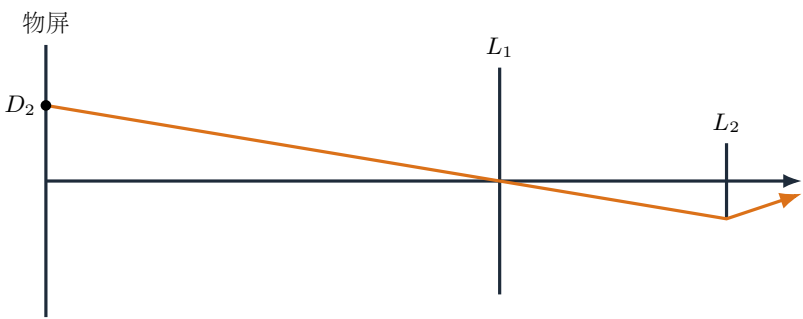
撞壁计数与冲量使用不同积分权重


l_1
l_2
物屏
L_1, D_1
L_2, D_2



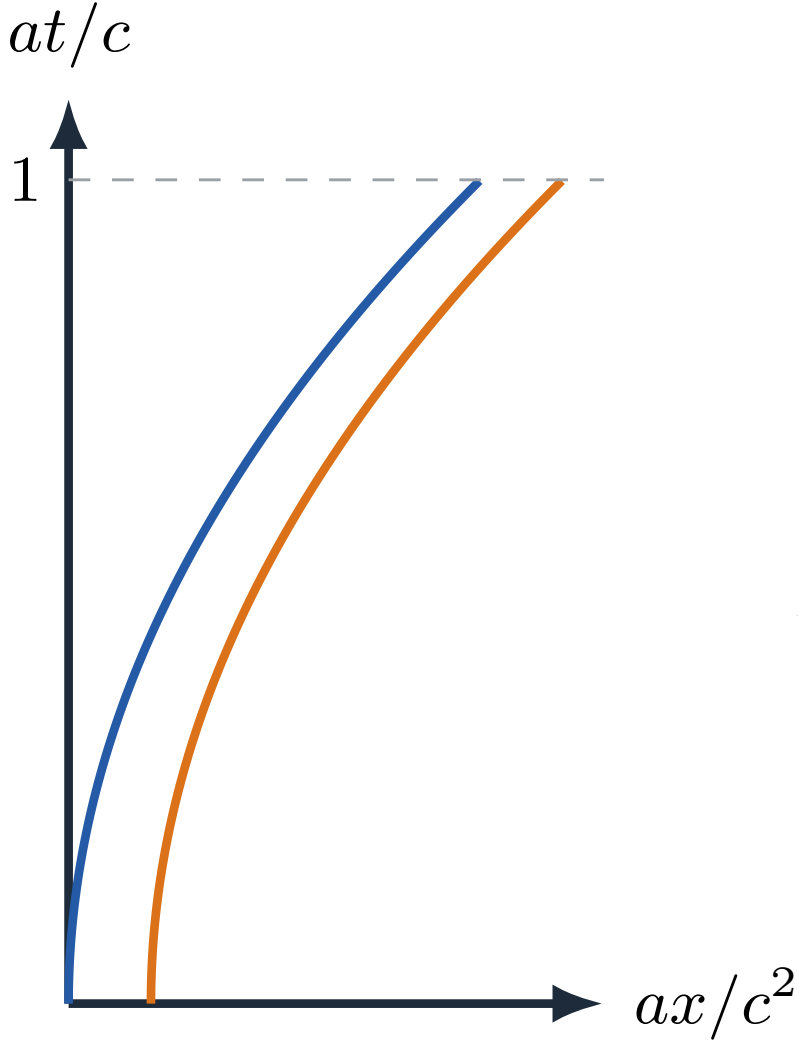
极限可见光线：$y_1 = -3D_2/2,\ y_2 = -D_2/2$

存在任意一条可通过的光线



指定中心光线：$y_1 = 0,\ y_2 = -D_2/2$

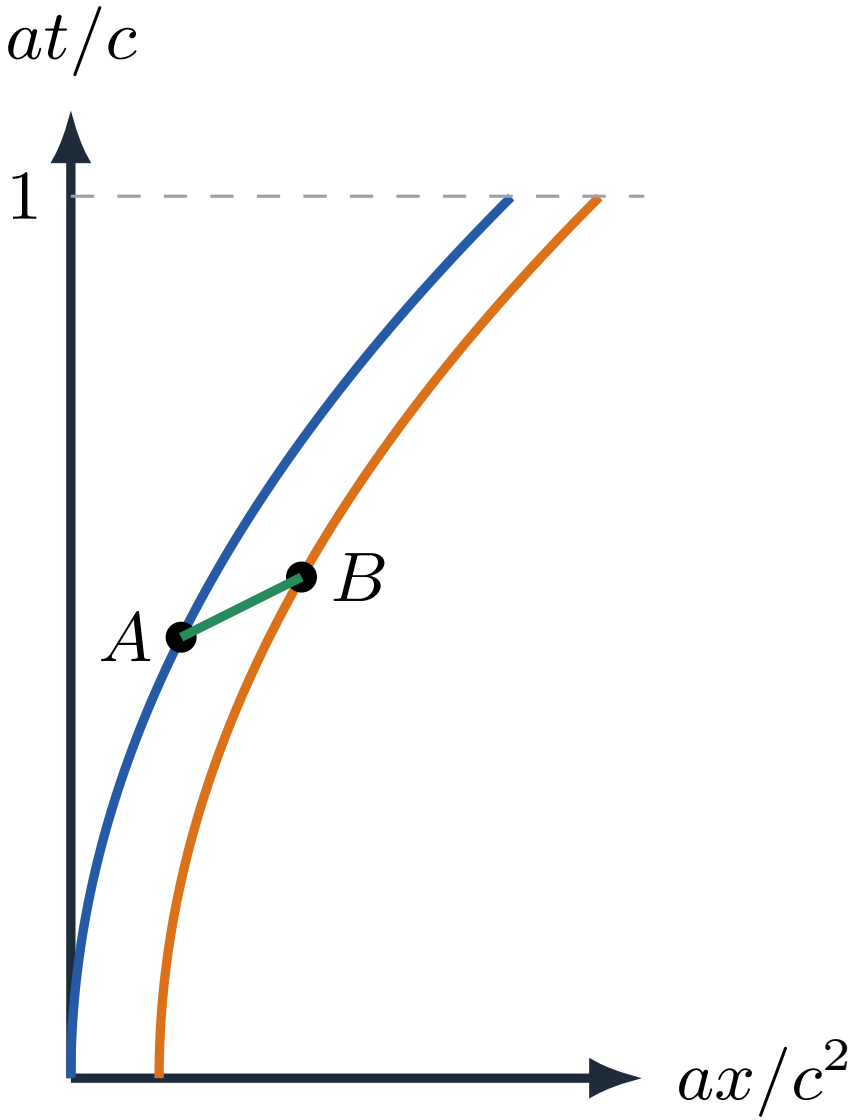
x 按 f_2、y 按 D_2 归一化，纵横比例不同



两端始终具有相同的地面坐标加速度

蓝：左端，橙：右端

只定义 $at/c < 1$ 的世界线



蓝、橙：左、右端世界线

绿色：事件 A 的瞬时同时线

示例 $al/c^2 = 0.1,\ at_A/c = 0.5$

两事件均须满足 $at/c < 1$